

9.6 MW Tesla Megapack BESS – Full Project Plan

Site: 340 Wilkinson Rd, Calipatria, CA 92233

Technology: Tesla Megapack 2 XL

Phase Count: 1 (Single Phase)

Project Size: 9.6 MW / 19.3 MWh

Total Estimated Cost: ~\$9.94M

Federal Incentives: 50% ITC (30% Base + 10% Energy Community + 10% Domestic Content)

1. Project Overview

This project proposes a single-phase, utility-grade Battery Energy Storage System (BESS) using Tesla Megapack 2 XL units installed at 340 Wilkinson Rd, Calipatria, CA. The system is designed to meet the minimum threshold utilities consider meaningful (approximately 10 MW), enabling participation in Resource Adequacy (RA), capacity markets, and grid-support services.

The project is structured to maximize federal incentives, minimize integration complexity, and position the asset for long-term refinancing after commercial operation.

2. Site Summary – 340 Wilkinson Rd, Calipatria, CA

- Flat, developable land suitable for energy infrastructure
- Agricultural zoning compatible with BESS development
- Located within an Energy Community, qualifying for an additional 10% ITC
- Proximity to Imperial Irrigation District (IID) grid infrastructure
- Estimated land control cost: \$20,000 (option agreement)



3. Technology – Tesla Megapack 2 XL

Tesla Megapack provides a fully integrated, factory-tested energy storage solution including:

- Battery modules and racks
- PCS/inverters
- HVAC and thermal management
- Fire suppression
- Battery Management System (BMS)
- SCADA/EMS controls
- UL-listed, CAISO-approved hardware

This integration significantly reduces EPC complexity and project risk.

4. System Configuration

- **Megapacks:** 5 units
- **Total Power:** 9.6 MW
- **Total Energy:** 19.3 MWh
- **Duration:** 2 hours
- **Interconnection Voltage:** 34.5 kV (typical IID feeder voltage)
- **Footprint:** ~1.5–2 acres

5. Costed Bill of Materials (External Items Only)

Tesla Megapack integrates all internal systems. The following BOM includes only external hardware and site-level components required for a complete, utility-grade installation. Refer table

Total Project Cost:

\$9,940,400

9.6MW BESS BOM

Category	Item	Qty	Unit Cost (USD)	Subtotal (USD)	Notes
Tesla Hardware	Megapack 2 XL (9.6 MW / 19.3 MWh)	5 units	\$5,045,000	\$5,045,000	From Tesla design tool
Medium Voltage Interface	MV Transformer (10–12 MVA, 34.5 kV)	1	\$600,000	\$600,000	Step-up transformer
	MV Switchgear lineup	1	\$900,000	\$900,000	Breakers, relays, metering
	Protection relays (SEL)	3	\$20,000	\$60,000	Feeder + transformer + PCS
	Utility revenue-grade meter	1	\$25,000	\$25,000	Required for billing
Civil & Structural	Megapack concrete pads	5	\$40,000	\$200,000	Tesla requires engineered pads
	Transformer/switchgear pad	1	\$60,000	\$60,000	Heavy pad
	Trenching + conduit	1 lot	\$250,000	\$250,000	LV + MV + control
	Access road + grading	1 lot	\$150,000	\$150,000	Gravel + drainage
	Fencing + gates	1 lot	\$80,000	\$80,000	Security perimeter
Cabling & Electrical	LV cabling (Megapack → transformer)	1 lot	\$120,000	\$120,000	AC cabling
	MV cabling (transformer → POI)	1 lot	\$180,000	\$180,000	MV cable + terminations
	Grounding grid	1 lot	\$70,000	\$70,000	Ground rods + bonding
	Auxiliary power panel	1	\$30,000	\$30,000	For site lighting, comms
Safety & Compliance	Site fire system (external only)	1 lot	\$50,000	\$50,000	External extinguishers, signage
	Emergency stop stations	4	\$2,500	\$10,000	Required by AHJ
Engineering & Soft Costs	Civil + electrical engineering	1 lot	\$400,000	\$400,000	Stamped drawings
	Permitting + inspections	1 lot	\$150,000	\$150,000	AHJ + utility
	Interconnection studies	1 lot	\$1,200,000	\$1,200,000	Utility-driven upgrades
	EPC labor + overhead	1 lot	\$1,000,000	\$1,000,000	Construction + integration
	Contingency (10%)	1 lot	\$500,000	\$500,000	Standard
Land	Land control (option/lease)	1	\$120,000	\$120,000	Estimated 12 Acres @10k/acre
Total				\$11,200,000	

6. Federal Incentives (50% ITC)

The project qualifies for:

- **30% Base ITC**
- **10% Energy Community ITC**
- **10% Domestic Content ITC**

Total ITC = 50%

On a ~\$10M project: **\$10M × 50% = \$5M ITC**

Bonus depreciation (60%) applies to the remaining basis.

7. Development Timeline

Months 0–2

- Land option signed
- Tesla reservation placed (\$10k)
- Preliminary engineering
- Interconnection application submitted

Months 2–6

- Utility studies
- Load flow, short-circuit, and protection review

Months 6–9

- EPC contracting
- Permitting
- Site civil preparation

Months 9–18

- Megapack delivery



- Installation
- Commissioning
- Permission to Operate (PTO)

8. Financing Strategy

A 10% equity strategy is viable:

- **Equity:** ~\$1M
- **Debt:** ~\$6–7M (construction loan)
- **Vendor terms:** 30–60 days typical
- **Refinance after COD:** 20–25 year amortization
- **Revenue:** RA + capacity + ancillary services

Interconnection approval is the key milestone that unlocks bank financing.

9. Summary

This project plan outlines a single-phase, utility-grade 9.6 MW Tesla Megapack BESS installation at 340 Wilkinson Rd, Calipatria, CA. The project is structured to maximize federal incentives, minimize integration complexity, and position the asset for long-term financial performance.

The total project cost aligns closely with the \$10M target, and the system qualifies for a 50% ITC, significantly improving project economics.

This plan is ready for utility engagement, EPC discussions, and financial modeling.